

Introduction:

In modern **e-commerce systems**, managing product availability notifications for a large number of users can become complex and inefficient. In this project, we focus on an online store system where many users are interested in products that may become unavailable temporarily. For example, a product can be out of stock, and hundreds of users may want to know immediately when it becomes available again. Using traditional object-oriented approaches, the system may repeatedly check product status or create tight connections between products and users, leading to inefficient communication and difficult system maintenance.

To solve this issue, the **Observer Design Pattern** is applied. This pattern allows users to subscribe to a product and automatically receive notifications whenever the product status changes. The product acts as the subject, while users act as observers waiting for updates. When the product becomes available again, all subscribed users are notified instantly without requiring manual checking. This approach improves system organization, reduces unnecessary operations, and provides a flexible and efficient notification mechanism within the e-commerce environment.

Problem Analysis:

In the online store system, many users may be interested in the same product, especially when the product is out of stock. For example, hundreds of users may want to know immediately when a product becomes available again. Using the traditional approach, the system may require users to repeatedly check the product status manually, which creates unnecessary operations and poor user experience.

In addition, directly connecting every user with the product in a tightly coupled way makes the system more difficult to manage and maintain. As the number of users increases, sending updates individually using traditional methods can reduce system efficiency and make the notification process more complex. Updating or modifying the notification mechanism may also require changes in multiple parts of the system.

Therefore, an efficient solution is needed to automatically notify all interested users whenever the product status changes, while keeping the system flexible and organized.

To solve these challenges, the Observer Design Pattern provides an effective solution by allowing users to subscribe to product updates. The product maintains a list of subscribed users and automatically sends notifications when the product becomes available again.

This approach reduces unnecessary checking, improves communication between objects, and increases the flexibility and scalability of the e-commerce notification system.

Why Observer Pattern?

The Observer Design Pattern is used in this project **because** the online store system needs to notify many users when a product becomes available again. Multiple users may be interested in the same out-of-stock product, and manually checking product availability repeatedly is inefficient and inconvenient.

The Observer Pattern solves this problem by allowing users to subscribe to product notifications. When the product status changes and becomes available, the system automatically sends notifications to all subscribed users. This removes the need for manual checking and improves communication between the product and users.

In addition, the pattern reduces tight coupling between objects, improves system organization, and makes the notification system more flexible, maintainable, and scalable.